

Parth Mhaske

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EDUCATION

University of Maryland

College Park, MD

B.S. in Computer Science and Applied Mathematics (GPA: 3.7)

Expected Graduation: May 2028

- **Coursework:** Algorithms, Data Structures, Computer Systems, Data Science, Probability, Compilers & Functional Programming
- **Awards:** 2x AIME; Science Olympiad Nationals 3rd; FRC World Championship Autonomous Award

EXPERIENCE

Corsha

May 2026 - Aug 2026

Software Engineering Intern

Vienna, VA

- Built and tested a Go service that infers partner API versions at handshake in 80ms, removing manual setup across 6+ deployments.
- Architected a standalone real-time data-ingestion service with disk-backed queues, sustaining 10,000+ logs/sec at 0% data loss.
- Automated StrongSwan IPsec deployment via Kubernetes Helm, eliminating manual restarts during certificate rotation.

Nuntius (YC S25)

Nov 2025 - May 2026

Software Engineer

- Engineered 6 agentic environments with automatic evaluation pipelines to benchmark multi-step LLM reasoning.
- Built custom reward system for LLM tool-call correctness; surfaced 88% frontier-model failure on 100+ multi-turn benchmarks.
- Conducted penetration testing and failure analysis on AI models, uncovering edge cases that informed hardening strategies.

Smith Investment Fund

Oct 2025 - Present

Quantitative Developer

College Park, MD

- Built a C++ Crypto.com HFT engine with Boost.Beast and lock-free MPSC queues, with 110 ns p99 internal tick-to-order latency.
- Designed an event-driven StrategyBase that each strategy subclasses to emit signals, validated on a JSONL replay backtester.
- Engineered a thread-safe order book and pre-trade risk manager to validate order flow and gate execution in real time.

RESEARCH

Department of Astronomy, University of Maryland

Jan 2026 - Present

Undergraduate Researcher

College Park, MD

- Reduced MSTM dust-scattering compute time from hours to seconds (1000x speedup) using a Gaussian Process surrogate.
- Mapped polarization profile of interstellar comet 3I/ATLAS across 4 wavelength bands along tail and from core.

MASS Lab, University of Texas at Austin

Apr 2024 - Jan 2025

Research Assistant

Austin, TX

- Modeled UAV impact dynamics under stochastic touchdown; presented at AIAA Aviation 2026 (DOI: 10.2514/6.2026-4336).
- Derived a 16 mode hybrid system of differential equations for the UAV's roll, pitch, and vertical motion during impact.
- Built a C++/OpenMP Monte Carlo over 5,000 stiffness-damping configs, sizing a suspension to a 25% structural force margin.

PROJECTS

NJ High Injury Network Generator | [Link](#) | *Python, FastAPI, PostgreSQL/PostGIS*

- Built a full-stack analytics and decision-support tool to identify statistically significant high-crash corridors in NJ.
- Integrated NJDOT crashes with OpenStreetMap road geometries in PostGIS to aggregate crashes by segment.
- Flagged high-crash corridors across 568 NJ municipalities statewide by applying Poisson tests to 5 open data sources.

Clarify | *Python, FastAPI, Neo4j, Docker, OpenAI API*

- Developed a digital-forensics platform linking evidence across files and messages; runner-up at Bitcamp 2026.
- Engineered FastAPI ZIP ingestion across 7 file types, deferring Neo4j findings recomputation until batch completion.
- Implemented a thread-safe LLM circuit breaker using Python threading.Lock, with a 4-failure threshold and 60s cooldown.

TECHNICAL SKILLS

Languages: Python, C++, Go, Java, SQL, C, HTML, CSS, JavaScript, Rust

Libraries/Frameworks: FastAPI, NumPy, SciPy, pandas, Scikit-learn, TensorFlow, Boost.Beast

Developer Tools: Git, Docker, Kubernetes, CI/CD, Claude Code, OpenAI Codex, Cursor, Grafana, Jupyter

Technical Areas: Distributed Systems, Observability, Backend Systems, Data Pipelines, High-Performance Systems